



*Advanced AI-based dynamic Integration
of heterogeneous data for semantic
clusters formalisation in human-centric
intelligent enterprise systems*

Logistic ontology & Dataset generation & Evaluation

Mario Lezoche

Thomas Pallet

Tuesday 07th April 2026

Table of content

01

Introduction

02

**Ontology &
Dataset**

03

Evaluation

04

Conclusion

Introduction

IOF Core as the foundation of the ontology

<https://github.com/iofoundry/ontology/tree/master>

Usage of

- MaterialEntity
- Agent
- Process

All disjoint (mutally), process is not a physical object and agent not a location

Introduction

Built then classes for those 3 cores

MaterialEntity :

subclass

LogisticsItem (things moved/stored)

- Products
 - RawMaterial
 - FinishedProduct
 - SparePart

PhysicalLocation (things that contain/locate other things)

- Warehouse
 - StorageZone
 - Shelf
 - Bin
 - Dock

Agent :

- LogisticAgent
 - Worker
 - Robot

Process :

- LogisticsProcess
 - TransportProcess
 - StorageProcess
 - PickingProcess

RawMaterial, FinishedProduct, SparePart disjoint (mutually)

same for LogisticsItem, PhysicalLocation, LogisticAgent

Introduction

Defined six forward properties (moves, performedBy, storedIn, locatedIn, hasOrigin, hasDestination)
each with a declared inverse (isMovedBy, performs, hosts, contains, isOriginOf, isDestinationOf).

SKOS (Simple knowledge organization System)

Wanted to combined the OWL syntax I had for now and add SKOS

SKOS use for taxonomy, in my case, because I will discover family and subfamily characteristics when reading documentation, I need to add them.

To do so, using skos only for this part can be good I think, I read stuff about SKOS and I am not sure that it is a good idea but we will try it and if it is not good, then we will still use OWL but for the whole ontology and the taxonomy won't be as precise as what skos are but we will see

Table of content

01

Introduction

02

**Ontology &
Dataset**

03

Evaluation

04

Conclusion

Ontology creation

Performed an interview with the client (Gauthier) in 3 steps :

Phase 1:

- Can you describe this file and what these columns represent?

Phase 2:

- For each column, can you explain concretely what it represents?
- Are there relationships between some columns?
- How do family and subfamily work (taxonomy, classification logic)?
- What units are used (dimensions, weight)?
- What does annual_sale represent?
- What is the relationship between annual_sale and ABC_consumption?

Ontology creation

Phase 3:

- Is Reference a unique identifier?
- Can there be duplicates?
- Can multiple products have the same designation?
- Is the designation sufficient to understand the product?
- What is the exact relationship between family and subfamily?
- Do the values (e.g., 51, 13) have a meaning or are they arbitrary?
- Which columns should be interpreted together?
- Are there strong dependencies between some columns?
- Quantity: is it fixed or variable?
- Weight: what unit?
- Length / width / height: what units?
- How is annual_sale calculated?
- How is ABC_consumption determined?
- What is the exact relationship between annual_sale and ABC_consumption?
- Is it a fixed or evolving rule?
- What relationships are “obvious” to you but not present in the data?
- What is missing to fully understand a product?

Interview : transcript and translation

Teams, raw txt files, split it in 3 phases, then correct it and translate it in english

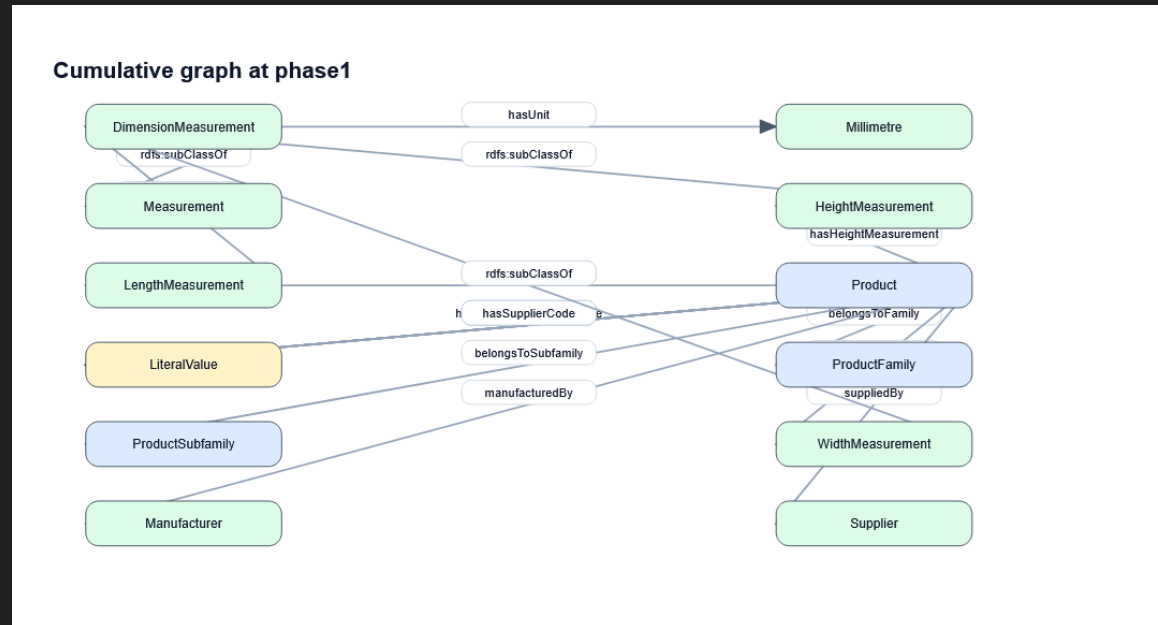
→ Verify that the ontology is complete, when parsing txt files to make sure correct relations are in the ontology structure

Two-level ontology design:

1. Schema-level : reading this documentation make changes like
(Dimension, hasUnit, Millimetre)
2. Instance-level : reading the CSV file populate the ontology
(Product_ABXDDF1044, hasLengthMm, 276.0)

Interview results

Automatic rule-based patterns to detect relations automatically, to make sure ontology was stable and that contains everything needed for the study.



Interview results

Problems : ambiguity in text like:

- Dimension should be in centimeter or millimeter, then what should we do about that ? My ontology can infer both of this, is this not okay ?
- Same with weight obviously, gram / kilogram
- Problem of unit, it says quantity = 4 but in fact it depends on 'conditionnement', like it can be quantity of 4 individual product or it can be 4 element of pack of 10 individual product !
→ Big problem because it depends every time, so we can't say anything about that.

⇒ Decision it to remove those parts because not relevant for now so that everything is clear !

Information of suppliercode in my ontology now, then a simple algorithm will parse all product_id and keep the supplier code and the article code for the instance-level knowledge.

Dataset generation

Based on the 3 phases, 117 sentences, 80 with mentions and 69 with at least one extractable candidate relation.

Generation based on those sentences of 100 documents :
1056 annotated sentences (one triple per sentence), and 1056 'INVALID' sentences

Final split :

- training: 1680 (80%)
- validation: 202 (10%)
- testing: 230 (10%)

Each example, initial sentence, annotated sentence, relation evaluated and the entities.

To do so, rule-based generated python file :

text, label, candidate_relation, subject, object, tier, and sentence.

label = words that make it so that the candidate relation is relevant

Table of content

01

Introduction

02

**Ontology &
Dataset**

03

Evaluation

04

Conclusion

Input format & tokenisation

Direct JSONL, sentence marked with token like [REL], [E1], [E2]

Tokenizer used from bert-base-uncased, addition of those 6 tokens:

[REL], [/REL], [E1], [/E1], [E2], [/E2]

→ treated as stable structural symbols from the tokenizer !

Padded length of 64 tokens (from my dataset, between 22-26) but very very long sentence won't be used, need to keep this part in mind

[REL] hasUnit [/REL] The [E1]dimensions[/E1] are expressed in [E2]millimetres[/E2].

[REL] hasUnit [/REL] The [E1]millimetres[/E1] are expressed in [E2]dimensions[/E2].

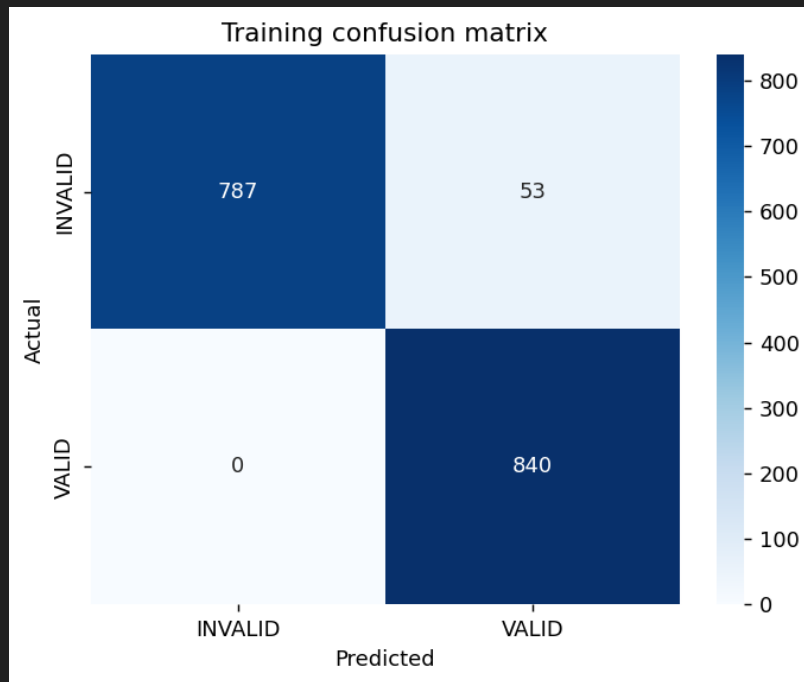
Model choice & training

Tested bert-base-uncased : binary relation classification (VALID / INVALID) (binary classification)

loaded from HF
(AutoModelForSequenceClassification)

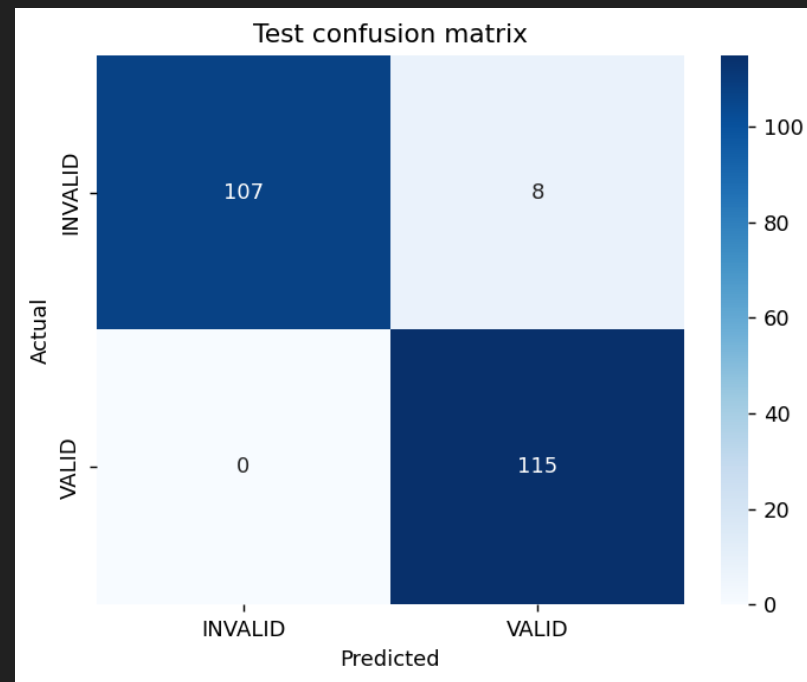
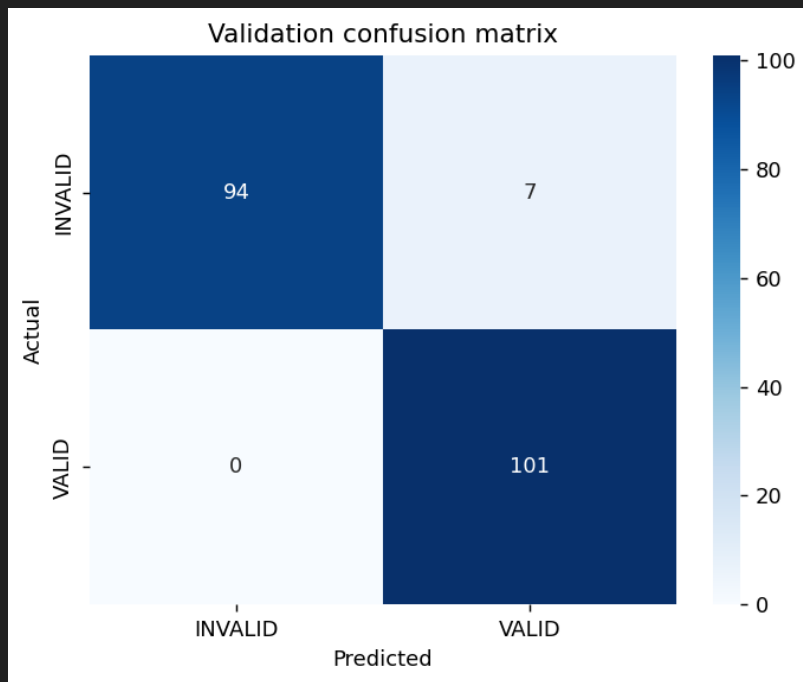
learning rate $2e-5$, EPOCHS = 3, batch_size = 32 for training, 64 for validation and test, AdamW optimizer.

Checkpoints are saved per epoch, and the best checkpoint is selected by validation accuracy.



Model evaluation

Confusion matrices for the validation and testing dataset.



Model evaluation

Recap

	Accuracy	Recall	Precision	F1-score
Training	96.8%	100%	94.1%	96.9%
Validation	96.5%	100%	93.5%	96.7%
Testing	96.5%	100%	93.5%	96.7%

Verifier learned the task effectively on this dataset, but this one is really structured, synthetic and generated from a controlled pipeline.

- + only single target triple

Model evaluation

Testing on more robust data, because here only based on the interview.

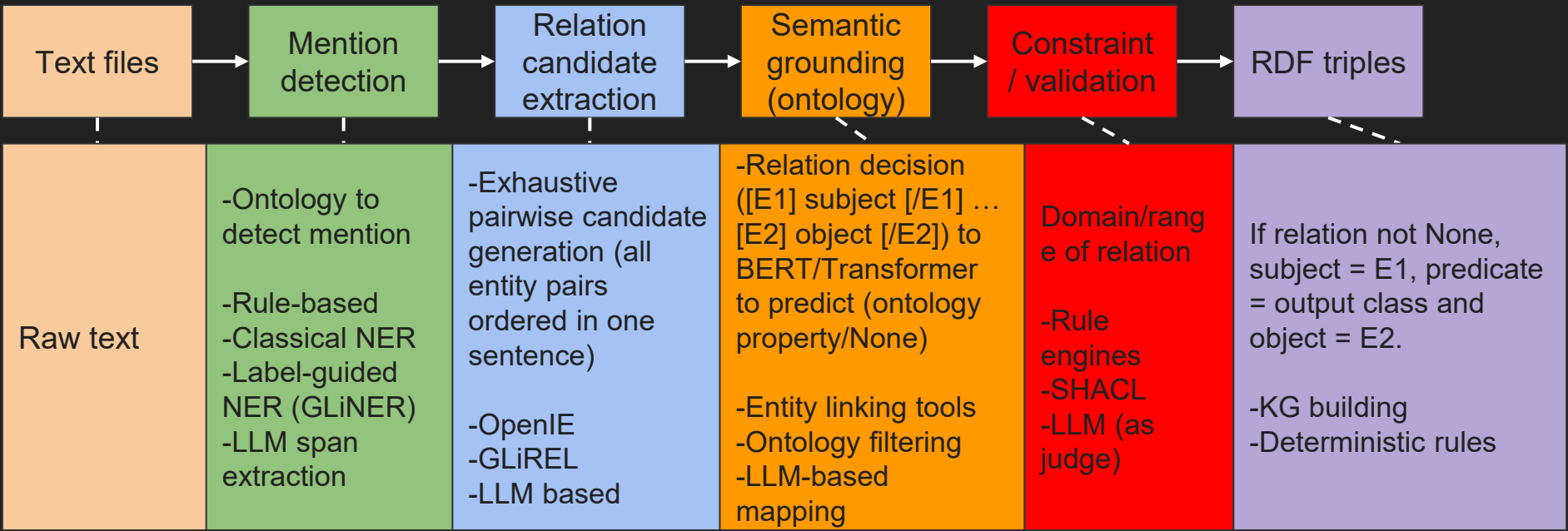


Table of content

01

Introduction

02

**Ontology &
Dataset**

03

Evaluation

04

Conclusion

Next steps

Writing of the article

→ need your review of it so I need to do that NOW

Talk about limitation in the article if that is okay ?

Thank you for your attention !

Questions ?